

A Low Power Content Addressable Memory Using Low Swing Search Lines

Byung-Do Yang, Yong-Kyu Lee, Si-Woo Sung, Jae-Joong Min, Jae-Mun Oh, and Hyeong-Ju Kang

Abstract—This paper proposes a low power content addressable memory (CAM) using low swing search lines. The CAM reduces the swing voltage and the power consumption of the search lines by using CAM cells as amplifiers. The CAM cells compare the stored data with the low swing search data on the search lines. The CAM also reduces the power consumption of match lines by using low swing NAND-NOR match lines. The 128×144 bit CAM chip was fabricated using a $0.18 \mu\text{m}$ CMOS process with $V_{DD} = 1.8 \text{ V}$. The CAM chip dissipates $2.82 \text{ fJ/bit/search}$ and consumes 8.7% of the power used by a conventional dynamic NOR-type CAM. It saves 83.9% and 97.3% of the power in the search lines and the match lines, respectively. Its area is 1.14 mm^2 . Its maximum operating frequency is 210 MHz .

Index Terms—Content addressable memory (CAM), low swing, match line, NAND cell, NOR cell, search line.

I. INTRODUCTION

CONTENT addressable memory (CAM) provides a fast data search function by comparing the search data with all of the stored data in a single clock cycle. Due to its high searching speed, CAM is a very attractive solution for a number of applications requiring fast searching functions, such as network routers, lookup tables, and data compressions. The fast searching using a fully parallel comparison method results in considerably large power consumption.

Fig. 1 shows the traditional CAM architecture, consisting of memory cells, a search word register, a word match circuit, and an address encoder. The CAM cell is implemented using NAND or NOR CAM cells. The memory cells in the CAM cells are made up of binary or ternary CAM memory cells. The binary CAM memory cell stores “0” or “1.” The ternary CAM memory cell stores “0,” “1,” or “X” (don’t care).

The search data on search lines (SLs) are compared with the stored data in all of the CAM memory cells in order to find

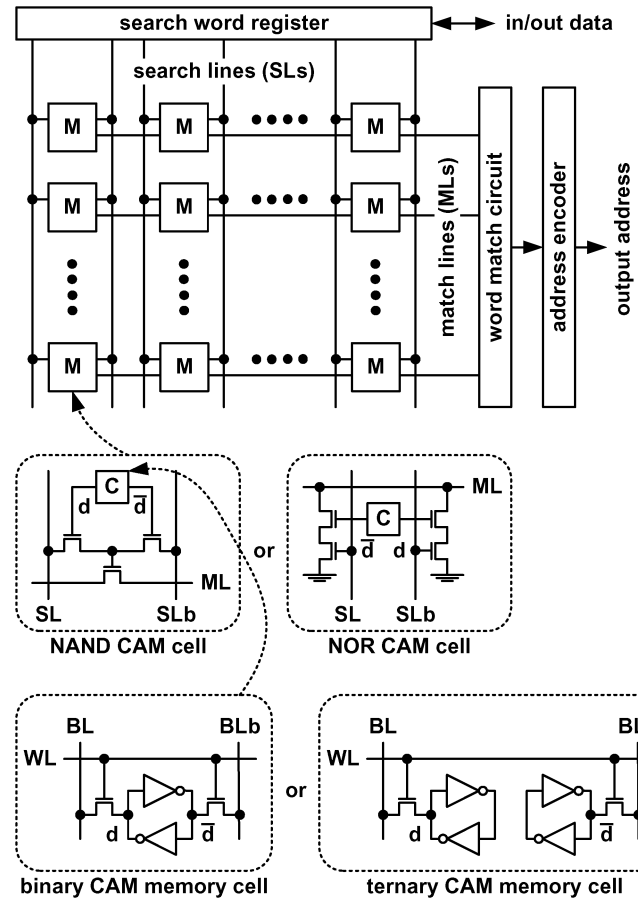


Fig. 1. Traditional CAM architecture.

a match line (ML) storing the search data. Most of the power used in the CAM is consumed by the highly capacitive SLs and MLs. In order to reduce the power consumption of the CAMs, many circuit techniques have been proposed. The NAND-type CAM consumes the least power in the MLs but it is slow, because only a precharged ML is discharged through a long series transistor chain [1]. In contrast, the NOR-type CAM is fast but it dissipates the largest power, because all precharged MLs except for a matched ML are discharged through parallel transistors [2]. In order to achieve both low power consumption and high speed, the NOR-type CAMs reduced the power consumption [3]–[14] and the NAND-type CAMs improved the searching speed [15]–[18].

Many techniques have been proposed to reduce the ML power consumption [2]–[18]. Most of them are based on the high-speed NOR-type CAM [3]–[14]. Some techniques reduced the ML power by allocating less power to mismatched MLs [4]–[7] or by lowering the swing voltage of the MLs using a

Manuscript received July 12, 2010; revised December 21, 2010 and March 21, 2011; accepted April 30, 2011. Date of publication July 14, 2011; date of current version November 29, 2011. This work was supported by the research grant of the Chungbuk National University in 2010. This was supported by the IT R&D program of Ministry of Knowledge Economy of Korea (MKE, 2008-F052-01, Development of next generation RFID technology for item level applications). This work was also supported by the Ministry of Knowledge Economy (MKE) and Korea Institute for Advancement of Technology (KIAT) through the Research and Development for Regional Industry (no. 70011159). This paper was recommended by Associate Editor M. Anis.

B.-D. Yang, S.-W. Sung, J.-J. Min, and J.-M. Oh are with Chungbuk National University, Cheong-Ju 361-763, Korea (e-mail: bdyang@cbnu.ac.kr).

Y.-K. Lee is with the Memory Division, Samsung Electronics, Yongin 449-900, Korea.

H.-J. Kang is with Korea University of Technology and Education, Cheonan 330-708, Korea.

Digital Object Identifier 10.1109/TCSI.2011.2158703

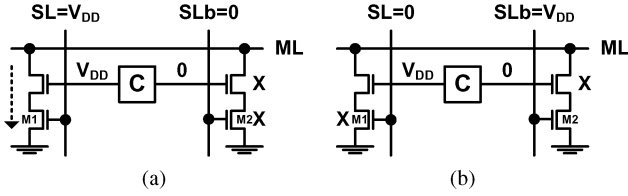


Fig. 2. Operation of a NOR-cell in a conventional CAM. (a) Mismatch. (b) Match.

charge-shared ML scheme [8], [9]. Other techniques reduced the number of activated MLs by using a pre-computation scheme [10] or a pipelined hierarchical search scheme [11]. However, they require additional circuits for the pre-computation registers or the pipeline registers. The pulsed NAND-NOR match-line scheme reduced the number of activated MLs without additional registers [12]. Some techniques are based on the low-power NAND-type CAM [15]–[18]. The AND-type ML scheme based on PF-CDPD logic achieved a fast searching speed [15]–[17]. The tree AND-type ML scheme improved the searching speed [18].

Several techniques have been proposed to reduce the SL power consumption [11]–[13], [16]–[18]. The pipelined hierarchical search scheme reduced the SL power by activating a few sub-SLs selected in the preceding pipeline stage [11]. However, this scheme has an increased latency and an area overhead due to the pipeline stages. The segmented SL scheme reduced the effective SL length by sorting all of the stored data with “X” (don’t care) cells and then blocking the signal propagation to the segmented SLs behind the “X” cells [13], [16], [17]. A drawback of this scheme is the difficulty in updating the stored data. If a word needs to be added, all of the stored data needs to be resorted and written at the system level. The charge-recycling SL driver reduced the SL power by recycling the charge on the SLs [12], [18].

In this paper, a low power CAM using low swing search lines (LSSL-CAM) is proposed. This method significantly reduces the SL power by comparing the stored data with the low swing search data on the SLs. It also further reduces the ML power by modifying the pulsed NAND-NOR CAM (PNN-CAM) architecture in [12].

Section II describes the architecture of the proposed LSSL-CAM. In Section III, we present the performance comparisons of the CAMs. Section IV shows the measurement results from the fabricated chip. This paper draws conclusions in Section V.

II. ARCHITECTURE

A. Concept of the Low Swing Search Line CAM

Fig. 2 shows the operation of a NOR-cell in a conventional NOR CAM when the search data on a SL pair (SL and SLb) is either mismatched or matched with the stored data. At first, the SLs are discharged to ground and the ML is precharged to V_{DD} . Then, either SL or SLb is charged to V_{DD} . In a mismatch case, the ML is discharged to ground. In a match case, the ML remains at V_{DD} .

Fig. 3 shows the operation of a NOR-cell in the proposed low swing search line CAM (LSSL-CAM). In order to compare the stored data with the low swing search data on a SL pair, an

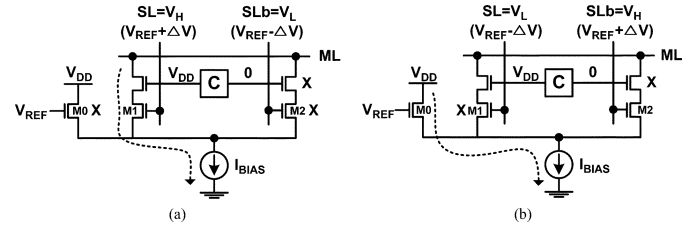


Fig. 3. Operation of a NOR-cell in the LSSL-CAM. (a) Mismatch. (b) Match.

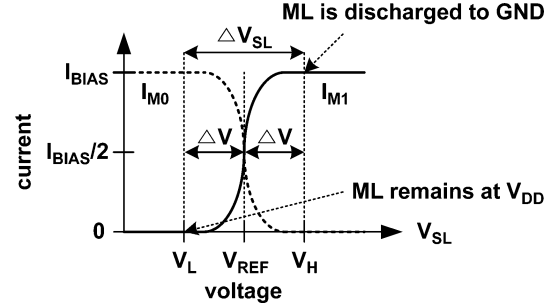


Fig. 4. The current variations in the LSSL-CAM versus V_{SL} .

NMOS transistor M0 and a bias current I_{BIAS} are needed. The gate voltage of M0 is V_{REF} . M0, M1, and I_{BIAS} act like a differential amplifier. The ML is discharged to ground or maintains at V_{DD} according to the voltages of SL and SLb. In a mismatch case, the voltages of SL and SLb are $V_H (= V_{REF} + \Delta V)$ and $V_L (= V_{REF} - \Delta V)$, respectively. The gate voltage of M1 (V_H) is ΔV higher than that of M0 (V_{REF}). Therefore, I_{BIAS} flows through M1. The ML is thereby discharged to ground. In a match case, the voltages of SL and SLb are V_L and V_H , respectively. I_{BIAS} flows through M0. The ML remains at V_{DD} .

Fig. 4 shows the current variations in M0 and M1 according to the SL voltage (V_{SL}). The currents in M0 and M1 (I_{M0} and I_{M1}) change according to V_{SL} . When $V_{SL} = V_H (= V_{REF} + \Delta V)$, $I_{M0} = 0$ and $I_{M1} = I_{BIAS}$. Conversely, when $V_{SL} = V_L (= V_{REF} - \Delta V)$, $I_{M0} = I_{BIAS}$ and $I_{M1} = 0$. ΔV has to be sufficient to make enough current difference between I_{M0} and I_{M1} . The SL swing voltage ($\Delta V_{SL} = 2 \times \Delta V$) in the LSSL-CAM is much smaller than the full swing voltage (V_{DD}) in the conventional CAMs. Therefore, the LSSL-CAM significantly reduces the SL power consumption.

Fig. 5 shows the operation of a NOR-cell block in the LSSL-CAM. M0 and I_{BIAS} are shared by all of the NOR-cells in the NOR-cell block. The area overhead due to M0 and I_{BIAS} is small. Before the search operation, the ML is precharged to V_{DD} . In a mismatch case, as shown in Fig. 5(a), K NOR-cells are mismatched. Cell current $I_{CELL} (= I_{BIAS}/K)$ flows in the mismatched NOR-cell. The total current flowing through the K mismatched NOR-cells equals I_{BIAS} . The mismatched ML is discharged by I_{BIAS} regardless of the number of mismatched NOR-cells (K). The ML sensing circuit detects the discharged ML. In a match case, as shown in Fig. 5(b), all of the NOR-cells are matched. I_{BIAS} flows through M0. The ML remains at V_{DD} .

Fig. 6 shows the schematic of the NOR-cell block in the LSSL-CAM. Fig. 7 shows the waveforms that illustrate the ML sensing operations of the NOR-cell block. The current supplying circuit generates I_{BIAS} . All of the bias currents in the NOR-cell blocks are mirrored by a current generation circuit. In order to reduce

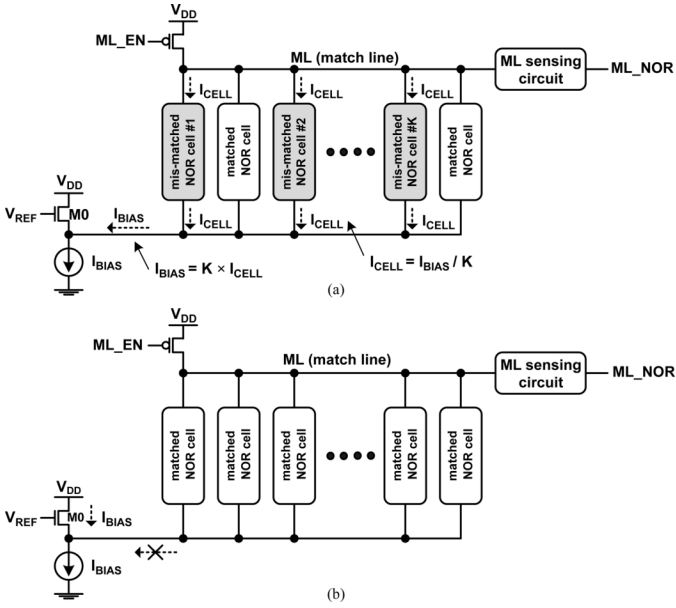


Fig. 5. A NOR-cell block in the LSSL-CAM. (a) Mismatch. (b) Match.

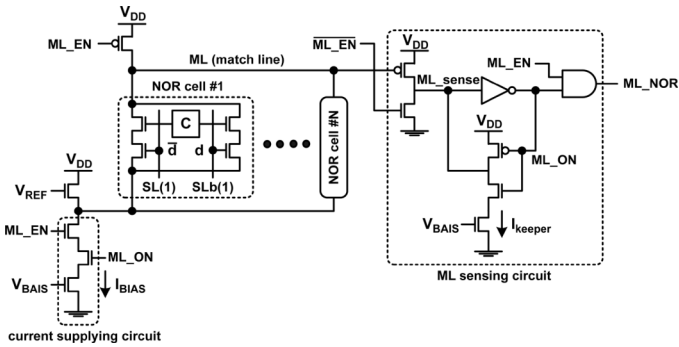


Fig. 6. Schematic of the NOR-cell block in the LSSL-CAM.

the ML power consumption, I_{BIAS} is dynamically controlled by the ML_EN (ML enable) and ML_ON (ML current on) signals. Initially, the ML_ON signal is “1.” After the SL voltages change, the ML_EN signal is set to “1.” This enables I_{BIAS} to flow. If all the NOR-cells are matched, the ML remains at V_{DD} . If not, the I_{BIAS} flowing through the mismatched NOR-cells decreases the ML voltage. The ML sensing circuit detects the voltage drop in the ML (ΔV_{ML}). When ΔV_{ML} becomes larger than the threshold voltage of the PMOS transistor, the ML_sense node is charged to V_{DD} and the ML_ON signal becomes “0.” A small amount of current I_{keeper} is used to decrease the ΔV_{ML} by reducing the fighting current in the ML_sense node. I_{keeper} is much smaller than I_{BIAS} . The ML_ON signal cuts the I_{BIAS} in the current supplying circuit. The I_{BIAS} control scheme reduces the power consumption in all of the mismatched MLs by limiting the ΔV_{ML} . Only a matched ML consumes the I_{BIAS} until the ML_EN signal returns to “0.”

B. Architecture of the LSSL-CAM

The architecture of the proposed LSSL-CAM is based on the pulsed NAND-NOR CAM (PNN-CAM) [12]. The LSSL-CAM uses the advantages of both the NAND CAM and the NOR CAM. The NAND CAM consumes a small amount of power but it is

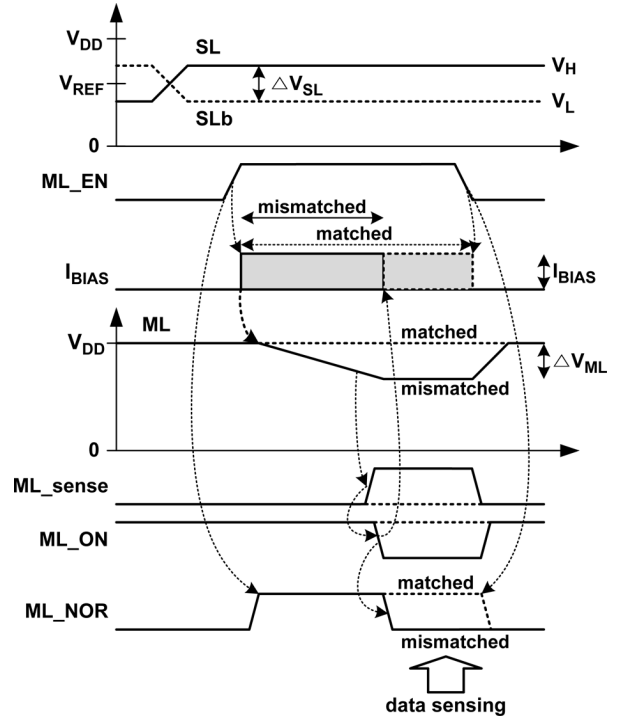


Fig. 7. Waveforms illustrating the ML sensing operations of the NOR-cell block.

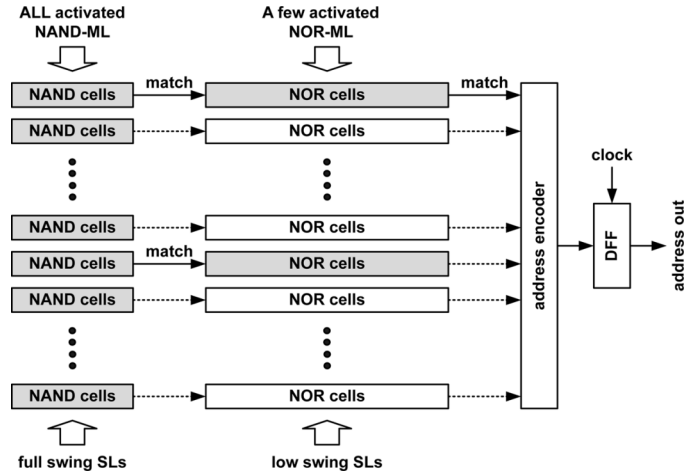


Fig. 8. Activations of the NAND-cell blocks and the NOR-cell blocks.

slow. Conversely, the NOR CAM is very fast but it consumes a great deal of power. The NAND-NOR architecture used in the LSSL-CAM reduces the number of activated NOR-cell blocks by using a few NAND-cells, as shown in Fig. 8. All of the NAND-cell blocks are activated but they consume a small amount of power.

The LSSL-CAM stores L words. Each word has $(K + N)$ bit data in a NAND-cell block (K NAND-cells) and in a NOR-cell block (N NOR-cells). If the matching probability of each bit is $1/2$, the matching probability of the K NAND-cells is $1/2^K$. The number of activated NOR-cell blocks is reduced from L to $L/2^K$. When $L = 128$ and $K = 8$, the average number of activated NOR-cell blocks is only 0.5.

The LSSL-CAM consumes a very small amount power in the MLs by using the NAND-NOR architecture. It further reduces the ML power by limiting the ML swing voltage (ΔV_{ML}). It uses a full swing voltage on a few of the SLs for the NAND-cells

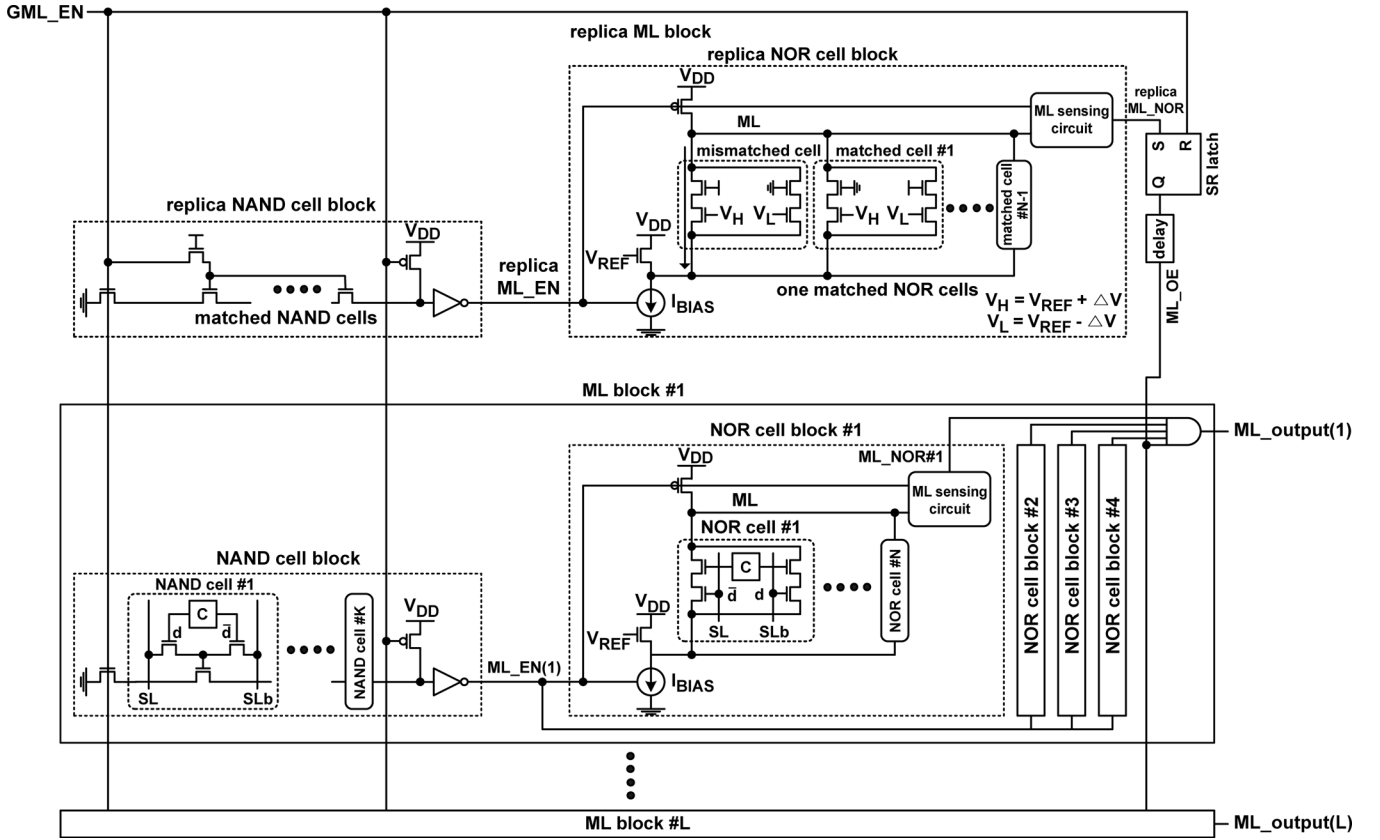


Fig. 9. Architecture of the LSSL-CAM.

and a low swing voltage on most of the SLs for the NOR-cells. Therefore, the SL power is significantly reduced.

Fig. 9 shows the architecture of the LSSL-CAM. It is divided into a replica ML-block and L ML-blocks. Each ML-block consists of a NAND-cell block and four NOR-cell blocks. The NAND-cell block and the NOR-cell block have K NAND-cells and N NOR-cells, respectively. Each ML-block has $K + 4 \times N$ CAM cells. To generate the control signals needed for the ML sensing operation, the LSSL-CAM utilizes the replica ML-block used in the PNN-CAM [12]. The delays of the GML_EN signal and the ML_OE signal are well matched to send the exact timing signals to all of the ML-blocks.

Fig. 10 shows the block diagram of the implemented LSSL-CAM with 128×144 bit ternary CAM (TCAM) memory cells. The LSSL-CAM was implemented using a $0.18 \mu\text{m}$ CMOS process with $V_{DD} = 1.8 \text{ V}$. Each ML-block has an 8 bit NAND-cell block and four 34 bit NOR-cell blocks. The I/O circuit drives the 144 SL pairs. The 8 SL pairs for the 8 bit NAND-cells use the full swing voltage ($V_{DD} = 1.8 \text{ V}$). The 136 SL pairs for the 136 bit NOR-cells use the low swing voltage ($\Delta V_{SL} = 0.4 \text{ V}$).

The three voltages for the low swing SL operation ($V_H = 1.2 \text{ V}$, $V_{REF} = 1.0 \text{ V}$, and $V_L = 0.8 \text{ V}$) are made by the SL voltage generator, shown in Fig. 11. When search data are changed, the voltages of some SLs are changed from V_H to V_L or from V_H to V_L . At this time, V_H decreases and V_L increases by the charge sharing between the SLs and the V_H and V_L nodes, respectively. The upper amplifier and the PMOS transistor keep V_H being higher than V_{ref_H} . The lower amplifier

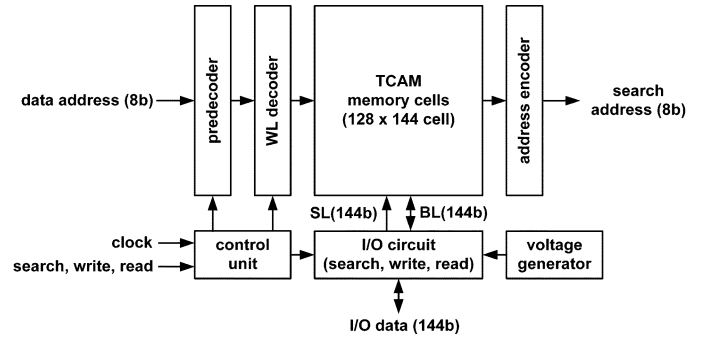


Fig. 10. Block diagram of the LSSL-CAM.

and the NMOS transistor keep V_L being lower than V_{ref_L} . Two amplifiers and a resistor string are designed to consume less than $20 \mu\text{A}$. In addition, they are only enabled during the search operation, in order to reduce a useless static power consumption.

Fig. 12 shows the simulated waveforms of the LSSL-CAM. The GML_EN (global ML enable) signal activates all of the NAND-cell blocks. A few matched NAND-cell blocks enable their ML_EN signals for NOR-cell blocks. Each ML_EN signal activates its bias current ($I_{BIAS} = 40 \mu\text{A}$). The bias current pulse length in the mismatch case is shorter than that in the match case. When the voltage drop in the mismatched ML becomes $\Delta V_{ML} (= 0.7 \text{ V})$, its I_{BIAS} is cut by the ML_ON signal from the ML sensing circuit. Before the ML_OE (ML output enable) pulse signal comes from the replica ML-block, all of the mismatched ML_NOR signals return to "0," which are the results of the NOR-cell blocks. The matched ML_NOR signal remains

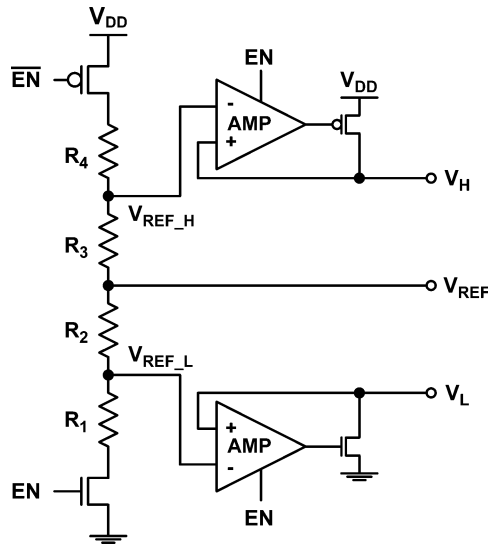


Fig. 11. SL voltage generator.

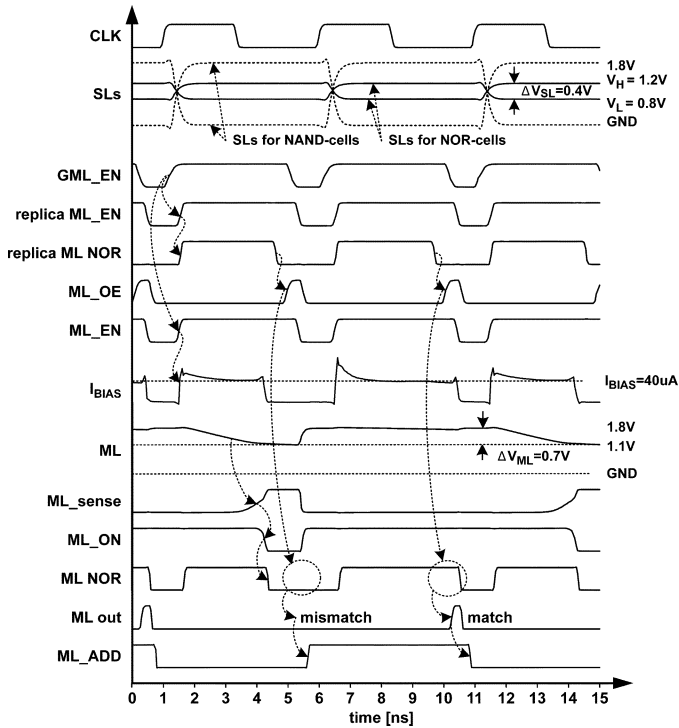


Fig. 12. Simulated waveforms of the LSSL-CAM.

at “1.” If the NAND-cell blocks are mismatched, their NOR-cell blocks are not activated and their ML_NOR signals remain at “0.”

C. Design Considerations

Fig. 13 shows the current flow in a 34 bit NOR-cell block. Fig. 14 shows the ML current variations according to $\Delta V (= \Delta V_{SL}/2)$ where $I_{BIAS} = 40 \mu A$, $V_{REF} = 1.0 V$, $V_L = V_{REF} - \Delta V$, and $V_H = V_{REF} + \Delta V$. The dotted lines show $\pm\sigma$ variations of the threshold voltage of the NMOS transistor in the NOR-cells.

In this simulation, 34 bit and 1 bit mismatch cases are included in order to obtain the best and worst delays of the 34 bit NOR-cell block. In the 34 bit mismatch case, most of the

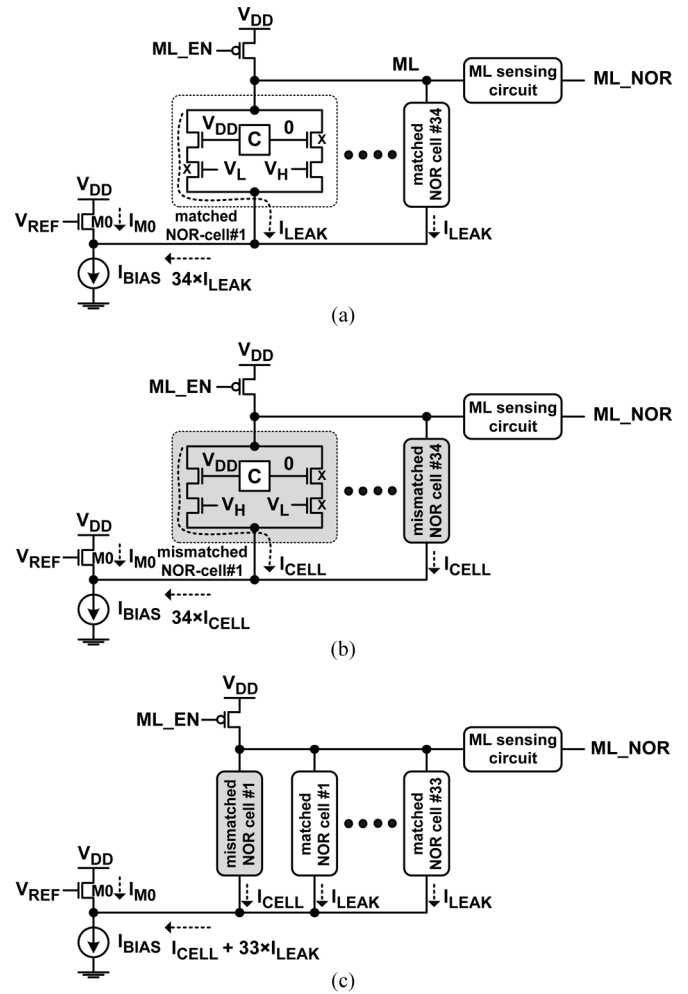


Fig. 13. Current flow in a 34 bit NOR-cell block. (a) Match. (b) 34 bit mismatch. (c) 1 bit mismatch.

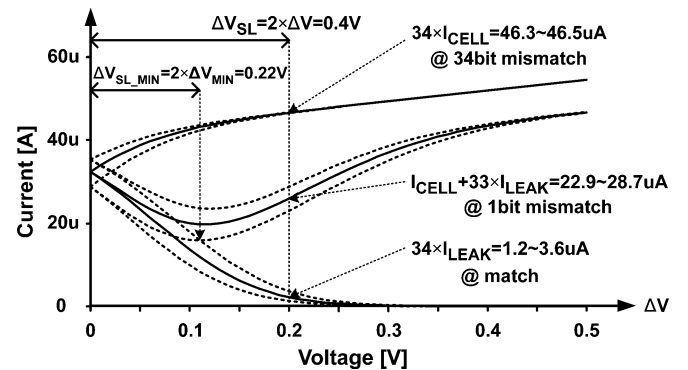


Fig. 14. ML current variations in a 34 bit NOR-cell block.

IBIAS flows through the 34 mismatched NOR-cells, which consists of 34 cell currents ($34 \times I_{\text{CELL}} = 46.3 \sim 46.5 \mu\text{A} @ \Delta V = 0.2 \text{ V}$). The current is a little larger than $40 \mu\text{A}$ because the I_{BIAS} is increased due to the channel length modulation effect in the bias current supplying circuit seen in Fig. 6. The ML voltage (V_{ML}) decreases to $V_{\text{DD}} - \Delta V_{\text{ML}} (= 1.1 \text{ V})$ and the ML is mismatched. In the 1 bit mismatch case, most of the I_{BIAS} flows through one mismatched NOR-cell ($1 \times I_{\text{CELL}}$). However, the mismatched NOR-cell does not conduct I_{BIAS} , due to the small size transistors used in the NOR-cell. A small amount

TABLE I
DELAYS IN THE LSSL-CAM

		Delay
8bit NAND-cell block		0.4ns
34bit NOR-cell block	1bit mismatched	2.8ns
	34bit mismatched	1.2ns
ML-block (1~34bit mismatched)		1.6~3.2ns
ML_OE signal (Replica ML block + delay)		4.0ns

of current flows through the 33 matched NOR-cells, which consists of 33 leakage currents ($33 \times I_{LEAK}$). The rest of the current flows through M0. The V_{ML} decreases by one cell current and 33 leakage currents ($1 \times I_{CELL} + 33 \times I_{LEAK} = 22.9 \sim 28.7 \mu A$ @ $\Delta V = 0.2 V$). The cell current difference according to the number of mismatched bits determines the delay variation in the 34 bit NOR-cell block. The best and worst delays of the 34 bit NOR-cell block are 1.2 ns and 2.8 ns in the 34 bit and 1 bit mismatch cases, respectively.

In the match case, most of the I_{BIAS} flows through M0, because the leakage current in the 34 matched NOR-cells ($34 \times I_{LEAK} = 1.2 \sim 3.6 \mu A$ @ $\Delta V = 0.2 V$) is very small. As the SL swing voltage ($\Delta V_{SL} = 2 \times \Delta V$) decreases, the leakage current increases, as shown in Fig. 14. The leakage current decreases the V_{ML} slowly. If the ML_OE pulse signal in Fig. 9 is generated before the V_{ML} decreases to $V_{DD} - \Delta V_{ML}$, then the ML sensing circuit decides that the ML is matched. The ML_OE pulse signal is made by the 1 bit mismatched replica ML-block and an additional delay for a timing margin, as shown in Table I. For the correct decision, the ML discharge current in the match case ($34 \times I_{LEAK}$) must be smaller than that in the 1 bit mismatch case ($1 \times I_{CELL} + 33 \times I_{LEAK}$), as shown in Fig. 14. Therefore, the minimum swing voltage (ΔV_{SL_MIN}) is 0.22 V. In this design, the ΔV_{SL} is selected to be 0.4 V.

In the conventional NOR-CAM [1], $V_H = V_{DD}$ and $V_L = 0 V$. V_H and V_L must be higher and lower than the threshold voltage (V_T) of the NMOS transistors in the NOR-cell in Fig. 2, respectively. If the delay requirement is ignored, the noise margins of V_H and V_L are $V_{DD} - V_T$ and V_T , respectively. However, the noise margins of the low swing search lines in the LSSL-CAM decrease. Fig. 15 shows the SL noise margins when $V_H = 1.2 V$, $V_{REF} = 1.0 V$, and $V_L = 0.8 V$. The noise margins of V_H and V_L (V_{H_NM} and V_{L_NM}) are 0.2 V and 0.17 V, respectively.

The bias current (I_{BIAS}) in the 34 bit NOR-cell block is determined by the delay and energy consumption, as shown in Fig. 16. The delay and energy consumption are measured using the 1 bit mismatch case. The energy consumption increases in proportion to the I_{BIAS} . However, the delay does not decrease in proportion to the I_{BIAS} because the I_{CELL} is limited by the small transistors in the NOR-cell. Therefore, I_{BIAS} is selected to be 40 μA .

Each ML storing a 144 bit word consists of an 8 bit NAND-cell block and four 34 bit NOR-cell blocks, as shown in Fig. 9. The bit allocation is chosen by the power and delay of the LSSL-CAM. Fig. 17 shows the power and delay comparisons according to the number of bits in a NAND-cell block (K). As K increases, the SL power increases, because the number of SLs using the full swing signals increases. However, as K increases, the ML

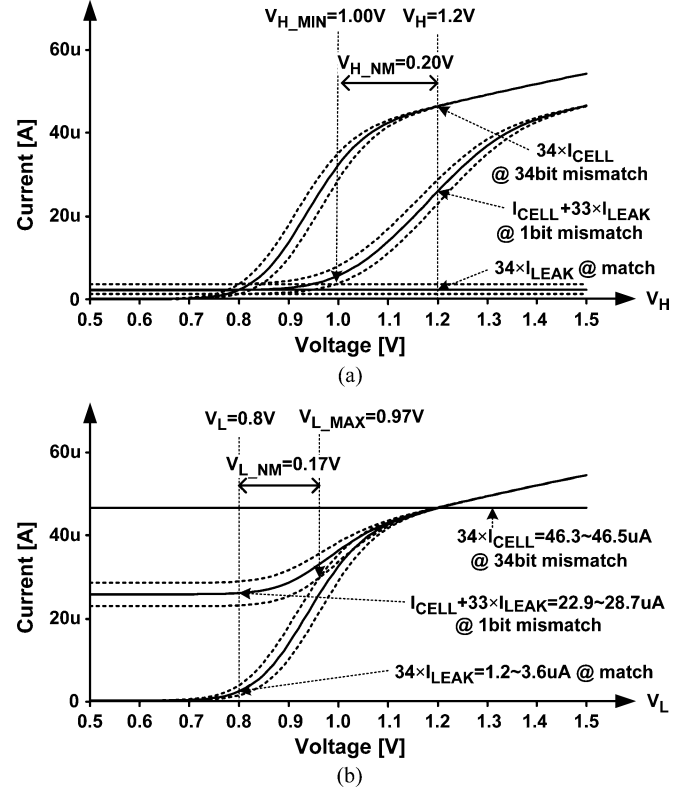


Fig. 15. Noise margins of: (a) V_H and (b) V_L .

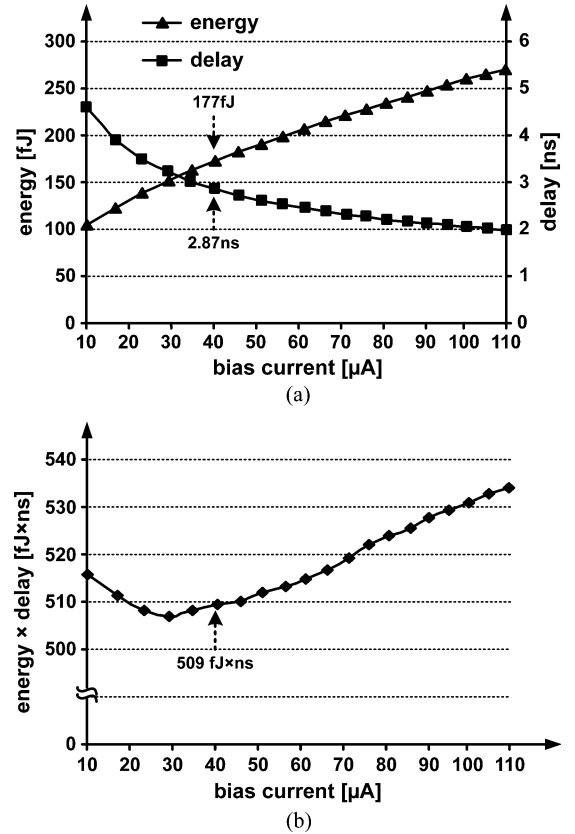


Fig. 16. (a) Energy and delay versus I_{BIAS} . (b) Energy delay product versus I_{BIAS} in the 34 bit NOR-cell block.

power significantly decreases, because the number of activated NOR-cell blocks is reduced from L to $L/2^K$, where L is the

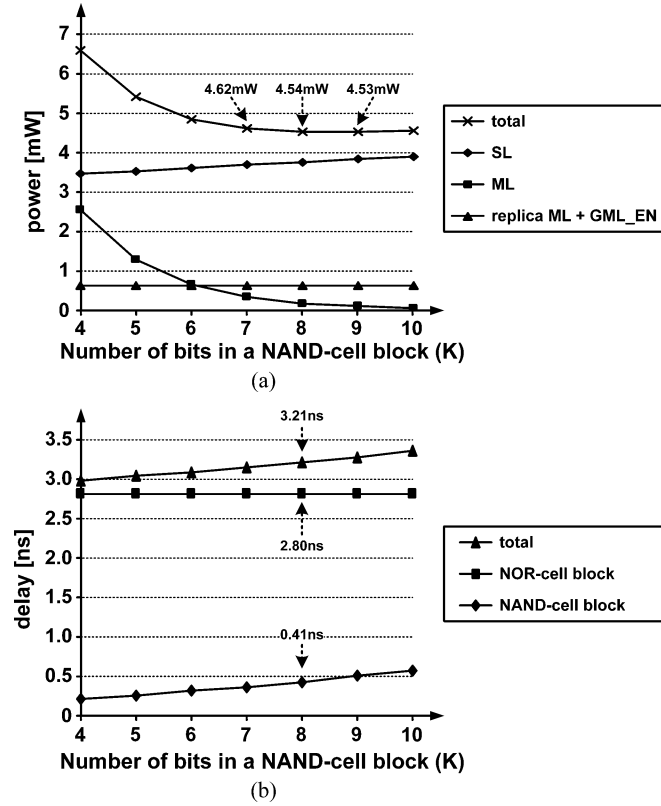


Fig. 17. (a) Power comparisons in the LSSL-CAM @ $f_{CLK} = 100$ MHz. (b) Delay comparisons in the NOR-cell block and the NAND-cell block.

number of MLs. The total power consumption, including all of the SLs and MLs, becomes the minimum at $K = 8 \sim 9$. The delay increases in proportion to K . Therefore, K is chosen to be 8.

Fig. 18 shows the energy, delay, and area comparisons of 136 bit NOR cells according to the number of bits in a NOR-cell block (N). The comparison results include 136 bit NOR cells, current supplying circuits, and ML sensing circuits. The number of NOR-cell blocks is therefore $136/N$. For example, when $N = 34$, the NOR cells are divided into 4 NOR-cell blocks. As N decreases, the delay decreases and the energy and area increase. The number of NOR-cell blocks is therefore chosen to be 4.

III. PERFORMANCE COMPARISONS

Table II shows the power consumption of the SLs. The precharged SL [1] consumes the largest amount of power, because the SL is precharged during every clock cycle. The non-precharged SL [5] consumes power only when its search data changes. The power consumption of the non-precharged SL is therefore less than that of the precharged SL. Generally, the transition probability of an SL (α) is lower than $1/2$. The charge recycling SL [12] further reduces the SL power consumption by recycling the charge in the SLs. However, it has power and area overheads due to its increased circuit complexity. The proposed low swing SL can significantly reduce the SL power. The SL power is reduced in proportion to the SL swing voltage (ΔV_{SL}). ΔV_{SL} can be reduced as long as the ML sensing is possible.

Fig. 19 shows the SL power comparisons for the NOR-cells. The power simulations are performed using a $0.18 \mu\text{m}$ CMOS

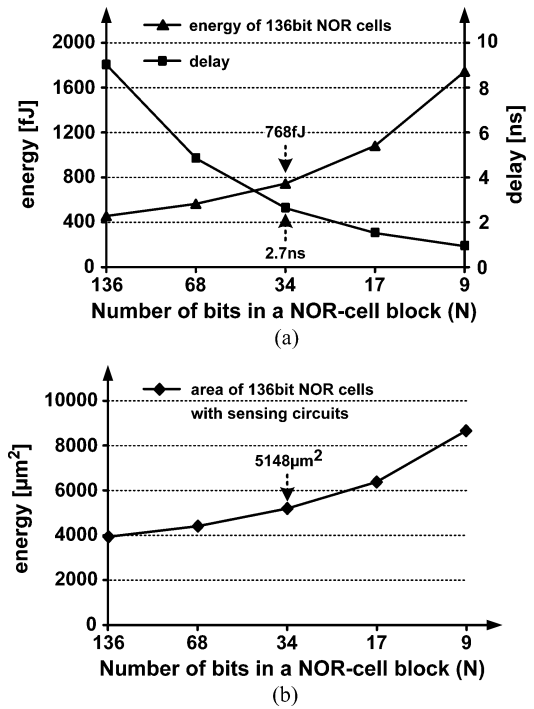


Fig. 18. (a) Energy and delay comparisons of the 136 bit NOR cells. (b) Area comparison of the 136 bit NOR cells.

TABLE II
SL POWER CONSUMPTION

	Power of an SL
Precharged SL [1]	$f \times C_{SL} \times V_{DD}^2$
Non-precharged SL [5]	$\alpha \times f \times C_{SL} \times V_{DD}^2$
Charge recycling SL [12]	$1/2 \times \alpha \times f \times C_{SL} \times V_{DD}^2$
Low swing SL	$\alpha \times f \times C_{SL} \times \Delta V_{SL} \times V_{DD}$

* α is the transition probability of an SL

process with $V_{DD} = 1.8$ V, $\Delta V_{SL} = 0.4$ V, and $f = 100$ MHz. The simulated results include the powers dissipated in the SLs and the SL driver circuits. Fig. 19(a) shows the SL power consumptions according to the number of words (L) when $\alpha = 1/2$. Fig. 19(b) shows the SL power consumptions according to α when $L = 128$. The low swing SL saves 86.2%, 71.7%, and 62.9% of the power compared to the precharged SL, non-precharged SL, and charge recycling SL, when $\alpha = 1/2$ and $L = 128$, respectively.

Table III shows the performance comparisons of CAMs with 128×144 bit TCAM cells. Fig. 20 shows the energy and delay comparisons of the CAMs. For the comparison, the dynamic NOR-type CAM (NOR-CAM) [1], the pulsed NAND-NOR CAM (PNN-CAM) [12], and the proposed LSSL-CAM are simulated using a $0.18 \mu\text{m}$ CMOS process with $V_{DD} = 1.8$ V. In the simulations, $\alpha = 1/2$, $\Delta V_{SL} = 0.4$ V, $V_H = 1.2$ V, $V_{REF} = 1.0$ V, and $V_L = 0.8$ V.

The NOR-CAM is fast but consumes a large amount of power. All of the MLs are precharged to V_{DD} and then all of the MLs except for the matched ML are discharged to ground. In addition, all of the SLs are precharged to ground and then some of them are selectively charged to V_{DD} according to the search data.

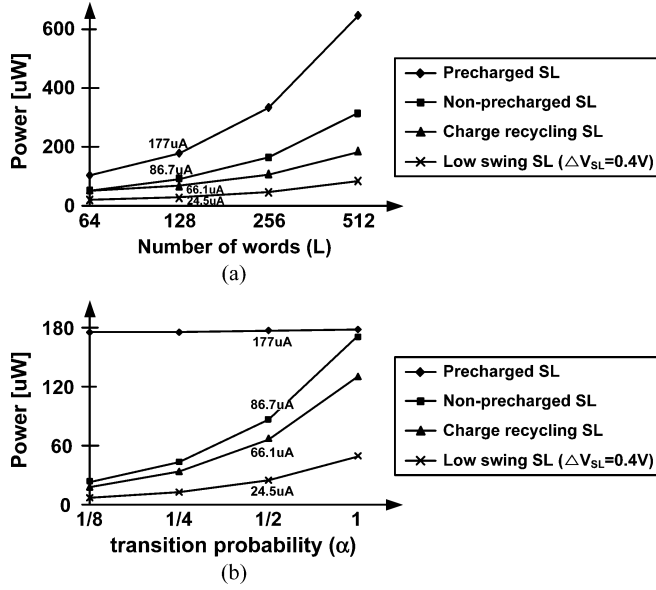


Fig. 19. SL power comparisons for the NOR-cells when: (a) the transition probability of an SL (α) is 1/2; (b) the number of words (L) is 128.

TABLE III
PERFORMANCE COMPARISONS OF CAMS

		NOR-CAM [1]	PNN-CAM [12]	LSSL-CAM
Technology		0.18 μ m CMOS		
Supply voltage		$V_{DD}=1.8$ V		
Organization		128 $\times$ 144bit ternary CAM		
ML	Total bit	144bits		
	ML (NAND)	-	4 $\times$ 6bits	8bits
	ML (NOR)	144bits	4 $\times$ 30bits	4 $\times$ 34bits
Swing voltage [V]	SL (NAND)	-	1.8	1.8
	SL (NOR)	1.8	1.8	0.4
	ML (NOR)	1.8	1.2	0.7
	ML (NAND)	-	1.8	1.8
Noise margins of SL [V]	V_H	0.72	0.72	0.20
	V_L	0.48	0.48	0.17
Search power [mW] @ $f_{CLK}=100$ MHz	SL	25.54	9.70	4.12
	ML	34.29	1.59	0.93
	Encoder + Control unit	0.15	0.15	0.15
	Total	59.98	11.44	5.20
Energy/bit/search [fJ]		32.54 (1.00)	6.21 (0.19)	2.82 (0.09)
Max. clock speed [MHz]		208 (1.00)	294 (1.41)	217 (1.04)
Chip core area [mm ²]		0.894 (1.00)	0.969 (1.08)	0.984 (1.10)

The PNN-CAM achieves both higher speed and lower power consumption than the NOR-CAM. Each ML with 144 bit cells is divided into four sub-ML blocks. The sub-ML block has 6 bit NAND-cells and 30 bit NOR-cells. The NAND-NOR architecture reduces the ML power. The charge recycling SL driver saves the SL power.

The proposed LSSL-CAM further reduces both the SL and ML power consumption. It reduces the SL power by using the low swing SL technique. It saves 83.9% and 57.5% of the SL power compared to the NOR-CAM and the PNN-CAM,

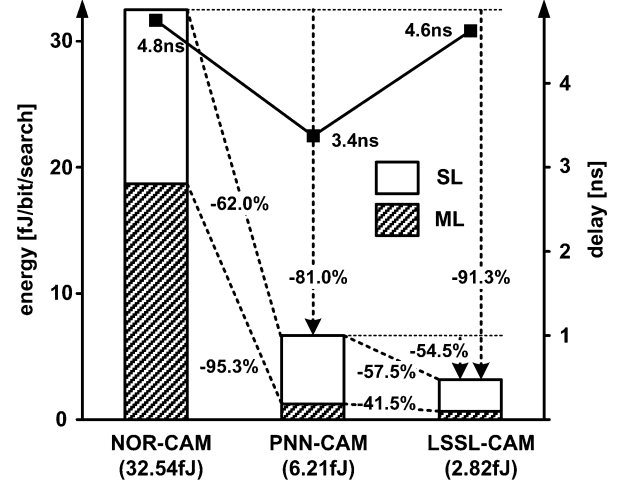


Fig. 20. Energy/bit/search and delay comparisons of the CAMs.

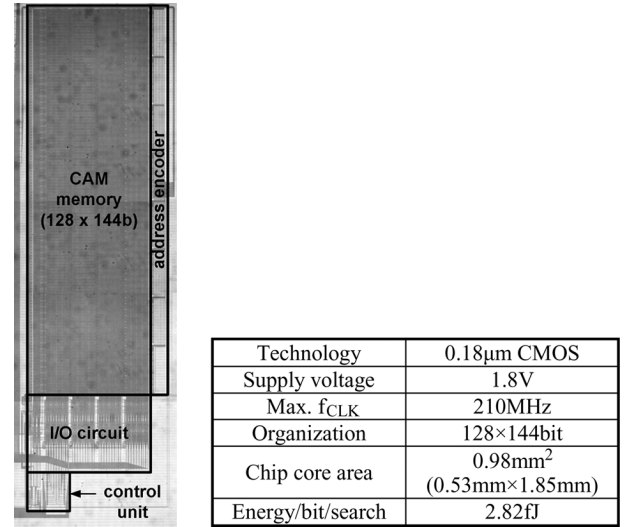


Fig. 21. Chip microphotograph and features of the LSSL-CAM chip.

respectively. It reduces the ML power by using the low swing ML sensing circuit. It saves 97.3% and 41.5% of the ML power compared to the NOR-CAM and the PNN-CAM, respectively. Although the LSSL-CAM is 35.3% slower than the PNN-CAM, it saves 91.3% and 54.5% of the power consumed by the NOR-CAM and the PNN-CAM, respectively.

IV. EXPERIMENTAL RESULTS

The chip microphotograph and features of the LSSL-CAM chip with 128 $\times$ 144 bit TCAM cells are shown in Fig. 21. This chip was fabricated using a 0.18 μ m CMOS process with $V_{DD} = 1.8$ V. The SL voltage generator was not included in the test chip. Its measured maximum clock frequency was 210 MHz. Its area is 1.14 mm². Its energy/bit/search is 2.82 fJ. Fig. 22 shows the waveforms of the LSSL-CAM chip at 200 MHz. Fig. 23 and Fig. 24 show the maximum clock frequencies according to the SL swing voltage (ΔV_{SL}) at $V_{DD} = 1.8$ V and the supply voltage (V_{DD}) at $\Delta V_{SL} = 0.4$ V. The results show that the LSSL-CAM works when ΔV_{SL} and V_{DD} are larger than 0.3 V and 1.2 V, respectively.

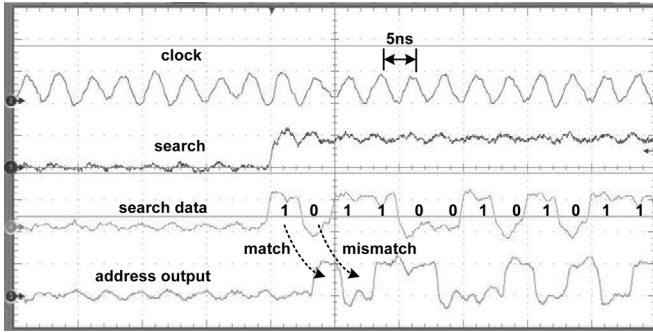
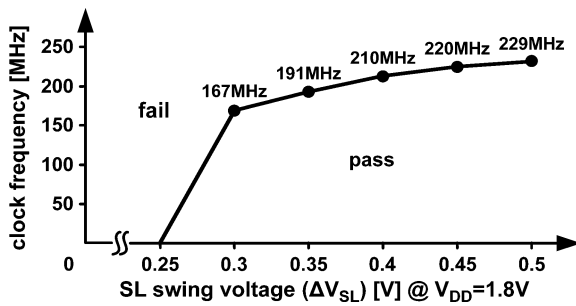
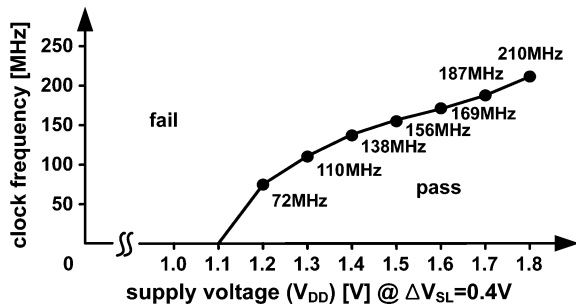


Fig. 22. Measured waveforms of the LSSL-CAM chip at 200 MHz.

Fig. 23. Max. clock frequency versus V_{SL} at $V_{DD} = 1.8$ V.Fig. 24. Max. clock frequency versus V_{DD} at $\Delta V_{SL} = 0.4$ V.

V. CONCLUSIONS

The LSSL-CAM using low swing search lines was proposed in this paper. The CAM cells in the LSSL-CAM act like amplifiers. They compare stored data with the low swing search data on the search lines. The LSSL-CAM significantly reduces the swing voltage and power consumption of the search lines. It also reduces the power consumption of the match lines by using the low swing NAND-NOR type match lines. The 128×144 bit CAM chip was fabricated in a $0.18 \mu\text{m}$ CMOS process with $V_{DD} = 1.8$ V. The chip core dissipates $2.82 \text{ fJ/bit/search}$. It consumes only 8.7% of the power used by the conventional dynamic NOR-type CAM. Its area is 1.14 mm^2 . Its maximum operating frequency is 210 MHz.

ACKNOWLEDGMENT

The chip fabrication was supported by the IC Design Education Center (IDEC).

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Byung-Do Yang received the B.S., M.S., and Ph.D. degrees in electrical engineering and computer science from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, in 1999, 2001, and 2005, respectively.

He was a Senior Engineer at the Memory Division, Samsung Electronics, Kyungki-Do, Korea, in 2005, where he was involved in the design of DRAM. Since 2006, he has been at Chungbuk National University, Korea. In 2008, he became an Assistant Professor. His research interests are memory circuits, analog

circuits, and power IC design.



Yong-Kyu Lee received the B.S. and M.S. degrees in semiconductor engineering from Chungbuk National University, Cheong-Ju, Korea, in 2008 and 2010, respectively.

Since 2010, he has been with the Memory Division, Samsung Electronics, Yongin, Korea, where he has been engaged in the design of PRAM. His research interests include low power and high performance PRAM.



Jae-Mun Oh received the B.E. degree in electronic engineering from Chungbuk National University, Cheong-Ju, Korea, in 2010. He is currently working toward the M.S. degree in semiconductor engineering from Chungbuk National University.

He is now working on boost converters for LED back light drivers and high efficiency buck converters for setup boxes. His research interests include high-efficiency power converter design.



Si-Woo Sung received the B.E. degree in electronic engineering from Chungbuk National University, Cheong-Ju, Korea, in 2010. He is currently working toward the M.S. degree in semiconductor engineering from Chungbuk National University.

His research interests include memory circuit design and organic-TFT circuit design.



Hyeong-Ju Kang (S'00–M'07) received the B.S., M.S., and Ph.D. degree in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, in 1998, 2000, and 2005, respectively.

From 2005 to 2006, he was with Magnachip Semiconductor, where he participated in the development of a smart card controller. From 2006 to 2009, he worked for GCT Semiconductor to develop baseband modem chips for several digital mobile TV standards.

Since 2009, he has been an Assistant Professor in the School of Computer Science and Engineering at the Korea University of Technology and Education, Cheonan. His current research interest includes communication modem design, embedded processor design, and computer-aided design algorithms for formal verification.



Jae-Joong Min received the B.E. degree in electronic engineering from Chungbuk National University, Cheong-Ju, Korea, in 2010. He is currently working toward the M.S. degree in semiconductor engineering from Chungbuk National University.

His research interests include data converter design and temperature sensor design.